

Designed for a shirt-sleeve environment, the 60- diameter cabin has a space-qualified Environmental Control System, as well as dual-pane windows. The ship uses three flight control systems – manual-subsonic, electric-supersonic and cold-gas RCS.

SpaceShipOne's hybrid rocket motor is a non-toxic, liquid nitrous-oxide/ rubber-fuel hybrid propulsion system. The avionics onboard provide the pilot with the precise guidance information needed to manually fly for boost and re-entry. It also provides guidance for approach and landing and vehicle health monitoring. The unit stores and telemeters flight test data to Mission Control.

The WhiteKnightOne is a manned, twin-turbojet research aircraft intended to provide a high-altitude airborne launch of SpaceShipOne. It is equipped with everything a spacecraft usually is, except rocket propulsion. The WhiteKnightOne's cockpit, avionics, ECS, pneumatics, trim servos, data system, and electrical system components are identical to those installed on SpaceShipOne.

There is an added benefit of the WhiteKnightOne in its ability to serve as a training vehicle. Its high thrust-to-weight ratio and enormous speed brakes allow the astronauts in training to practice space flight manoeuvres such as boost, approach, and landing with a very realistic environment. Thus, the aircraft serves as a high-fidelity moving-base simulator for SpaceShipOne pilot training.

Other White Knight mission capabilities include: reconnaissance, surveillance, atmospheric research, data relay, telecommunications, imaging & booster launch for micro-satellites.

SpaceShipTwo

The success of the SpaceShipOne led to several experts calling it the advent of private space flights. And sure enough, Scaled Composites found enough people knocking on their doors. Rutan eventually chose a partner in Sir Richard Branson and they together formed an alliance called The Spaceship Company for research purposes, while Branson would go on to form Virgin Galactic for commercial spaceflights in the future.

Together, they designed and built the SpaceShipTwo, a follow-up to the SpaceShipOne, which has not yet officially said when it will begin commercial spaceflights. Yet, over 400 tickets have already been booked at over \$200,000 per ticket.

SpaceShipTwo will be powered by a unique hybrid rocket motor, which is currently under development. The twin fuselage and central payload area